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Analysis of the Effect of Cold Treatment on the Mechanical Properties of Metallic Materials

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Cold treatment (down to −80 °C) and cryogenic treatment (down to the temperature of liquid nitrogen, −196 °C) are widely used in engineering as additional operations after quenching in order to improve the mechanical properties of steels. At the same time, no information has been found on the use of these types of treatment to improve the service properties of mining tools. This review analyses the literature data on the effect of cold treatment and cryogenic treatment on the properties of steels and discusses the possibility of applying them in mining machine building.

During the martensitic transformation that takes place on quenching, even when the steel is cooled to its M_f temperature, a certain amount of retained austenite remains in its structure; this amount increases if a steel whose M_f temperature lies below zero (Fig. 1) is cooled during quenching only to room temperature. The presence of retained austenite in the structure of steel lowers its hardness, strength and wear resistance, and also contributes to dimensional instability of parts [1].

As first proposed by A.P. Gulyaev in 1937, retained austenite in steel can be transformed into martensite by cooling the article to temperatures below room temperature, that is, by applying "cold treatment".

The effect of cold treatment depends on the amount of retained austenite in the quenched steel, which increases with increasing carbon content. Therefore, cold treatment is most effective for steels containing more than 0.4% carbon.

According to [2], cooling steel below −80 °C makes no sense, since it does not lead to additional martensitic transformation, which for most steels is completed above this temperature (Fig. 1). At the same time, research carried out in recent years has found [2] that continuing to cool martensite below the M_f temperature of the steel (for example, to the temperature of liquid nitrogen, −196 °C) together with a long holding time promotes the formation, during subsequent tempering of the steel, of a large number of very fine carbide particles, called η-carbide. As a result, the wear resistance and toughness of the material increase without a noticeable increase in its hardness. This type of treatment of steel has been named "cryogenic treatment".

[Figure 1 — see original PDF, page 3] Fig. 1. Dependence of the martensite start (M_s) and finish (M_f) temperatures on the carbon content in Fe–C alloys

The mechanism of the processes occurring during cryogenic treatment has not been sufficiently studied. It is possible that continued cooling to a low temperature increases the instability of martensite and changes its dislocation structure; carbon atoms in martensite migrate and cluster.

The process is slow, so holding must be carried out for a long time (24–72 hours). During subsequent holding at room temperature or during heating in the course of tempering, these fragments with an increased carbon concentration act as nuclei for the formation of fine carbide particles [2].

Cryogenic treatment has the most beneficial effect on steels with a minimum content of retained austenite (with a maximum content of martensite) in their structure, which indicates that this treatment acts precisely on martensite rather than on retained austenite. Its effect depends on the temperature and the holding time at the cryogenic temperature: the lower the temperature and the longer the holding time, the finer the η-carbide particles and the greater the increase in the wear resistance of the material.

[Figure 2 — see original PDF, page 4] Fig. 2. Effect of cold treatment and subsequent low-temperature tempering on the hardness of tool steel X12МФ (axes: hardness, HRC, versus austenitizing temperature, °C; curves: with cold treatment and without cold treatment)

The transformation of retained austenite in quenched steel into martensite is accompanied by an increase in stresses; therefore, during cold treatment articles must be cooled slowly, and tempering must be carried out immediately after the treatment [3]. Since holding steel after quenching at normal temperature for more than 3–6 hours stabilizes the austenite, cold treatment and cryogenic treatment must be carried out immediately after quenching. Thus, cold treatment and cryogenic treatment should be performed by uniformly cooling the specimens to the desired temperature, holding them at this temperature for 1–3 days, and then uniformly heating them to room temperature [4].

The structure of quenched steel with a large amount of retained austenite is not stable. Already at room temperature, and all the more so at low ambient (climatic) temperatures, retained austenite gradually transforms into martensite. This transformation is accompanied by an increase in volume and, as a consequence, by a change in the dimensions of the article. Cold treatment transforms retained austenite into martensite, which stabilizes the dimensions of the article. In many cases deep sub-zero cooling is not even required; it is sufficient to cool the quenched part in water at a temperature of +0 to +10 °C [1].

The effect of cold treatment on the hardness of steel as a function of the austenitizing temperature is illustrated in Fig. 2. In steels austenitized at high temperatures, the amount of retained austenite in the as-quenched state is greater than in steels austenitized at a low temperature and, as a consequence, the effect of cold treatment is stronger.

The dependences of the toughness of quenched steel austenitized at various temperatures on the temperature of cold treatment and cryogenic treatment are shown in Fig. 3, a.

It can be seen (Fig. 3, a) that, regardless of the austenitizing temperature, cold treatment of quenched steel (down to −70 °C) reduces its fracture energy for Charpy specimens by about 40% compared with similar specimens tempered twice at 200 °C after quenching. Carrying out the treatment at lower temperatures somewhat improves the toughness of the steel. The highest toughness of steel after cryogenic treatment is observed for specimens austenitized at lower temperatures (970 °C) and, as a consequence, containing the maximum amount of martensite in their structure. This is attributed [5] to microstructural changes in martensite that lead to the precipitation, during subsequent tempering of the steel, of the maximum number of fine (<5 μm) η-carbide particles (Fig. 3, b).

[Figure 3 — see original PDF, page 5] Fig. 3. Toughness of tool steel X12МФ (a) and the amount of carbide in its structure (b) after low-temperature tempering as a function of treatment temperature (axes: (a) fracture energy, J, and (b) amount of carbide, 1000/mm², versus cryogenic temperature, °C)

The results of experiments (the methodology is not reported) on the effect of these treatments on the wear resistance of steels are presented in the table [6].

Steel grade		Cold treatment at a temperature down to −79 °C	Cryogenic treatment at a temperature down to −190 °C
AISI	GOST	Improvement in wear resistance, %	
D2	Х12МФ	316	817
S7	6Х4М2ФС	241	503
52100	Х	195	420
O1	9ХВГ	221	418
A10	–	230	264
M1	Р2АМ9К5	145	225
H13	4Х5МФ1С	164	209
M2	Р6М5	117	203
T1	Р18	141	176
P20	38ХМ	123	130
440C	95Х18	128	121

Table 1

According to these data, the improvement in the wear resistance of steels resulting from cold treatment reaches a factor of 3, and that resulting from cryogenic treatment a factor of 8.

It can be concluded that cold treatment, like cryogenic treatment, can be used as an additional heat-treatment operation to increase the wear resistance of medium- and high-carbon steels used to manufacture the working tools of hydraulic breakers and other mining equipment operating under conditions of abrasive wear and significant impact loads.

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